

AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

BOA329

Ideal blueprint versus reality for CI/CD pipelines

Viktor Vedmich (he/him)

Senior Developer Advocate
AWS

Igor Ivaniuk (he/him)

Senior Solutions Architect
AWS



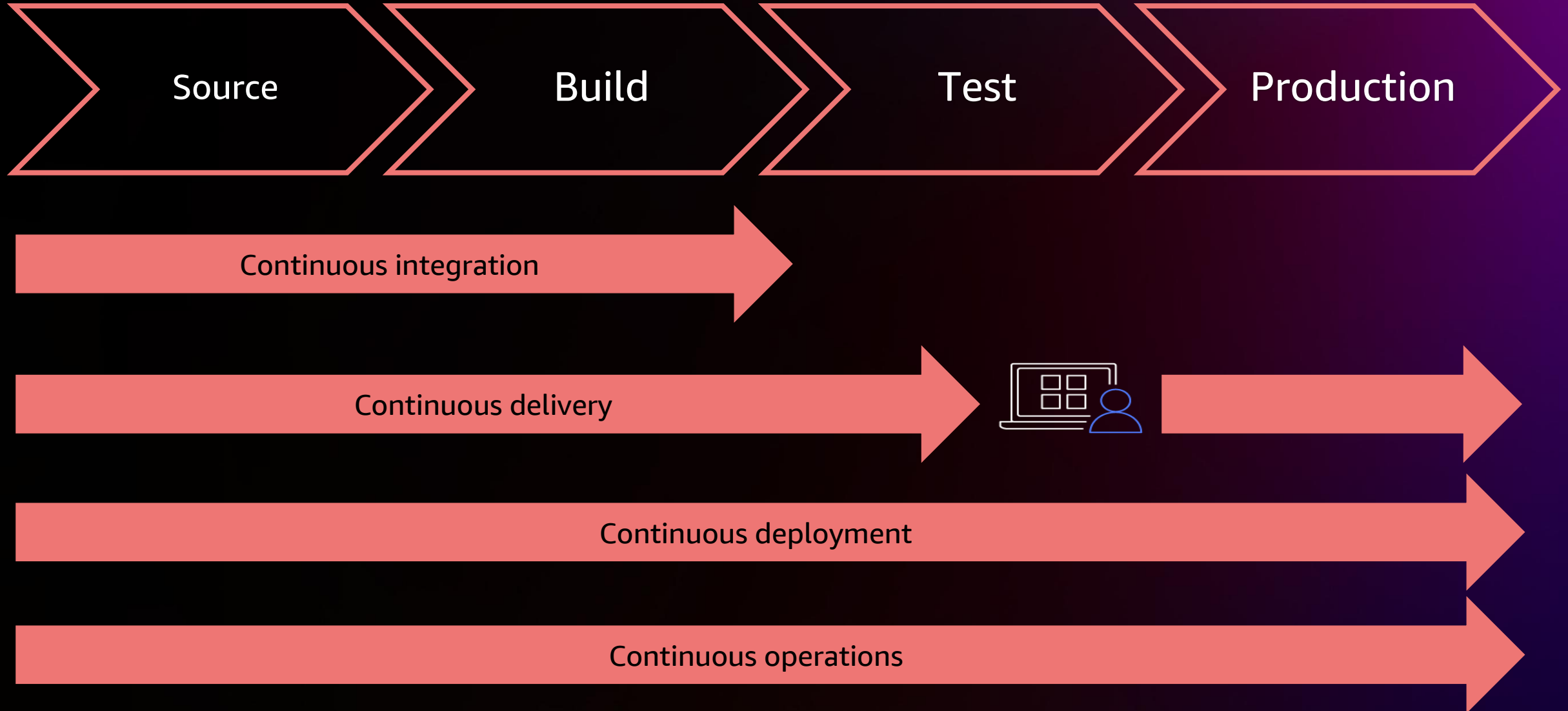
Agenda

- Release lifecycle
- Branching strategy
- CI/CD journey
- Quality gates
- Metrics
- Final blueprint
- Reality for CI/CD pipelines

Overview release lifecycle

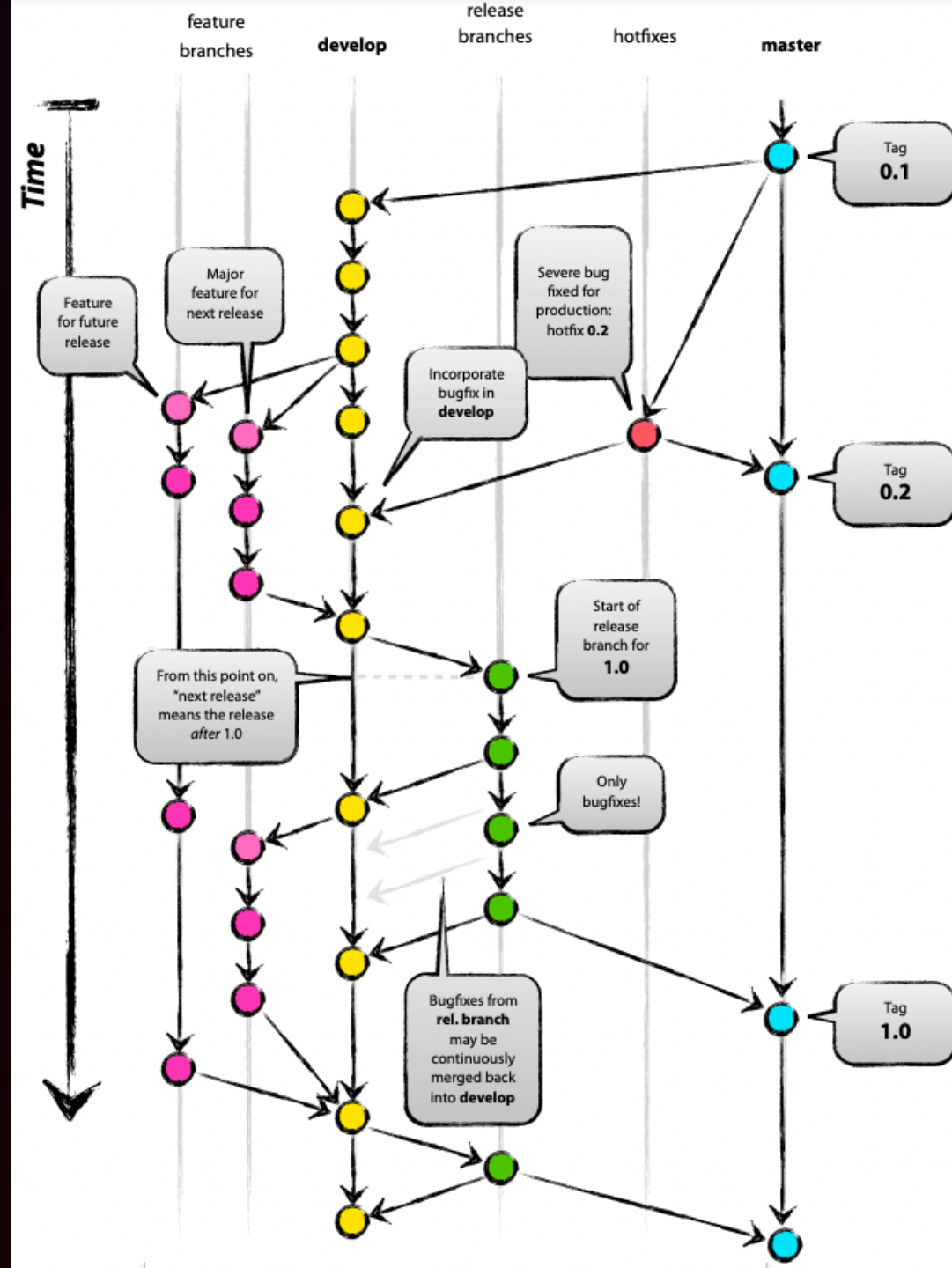


Release lifecycle



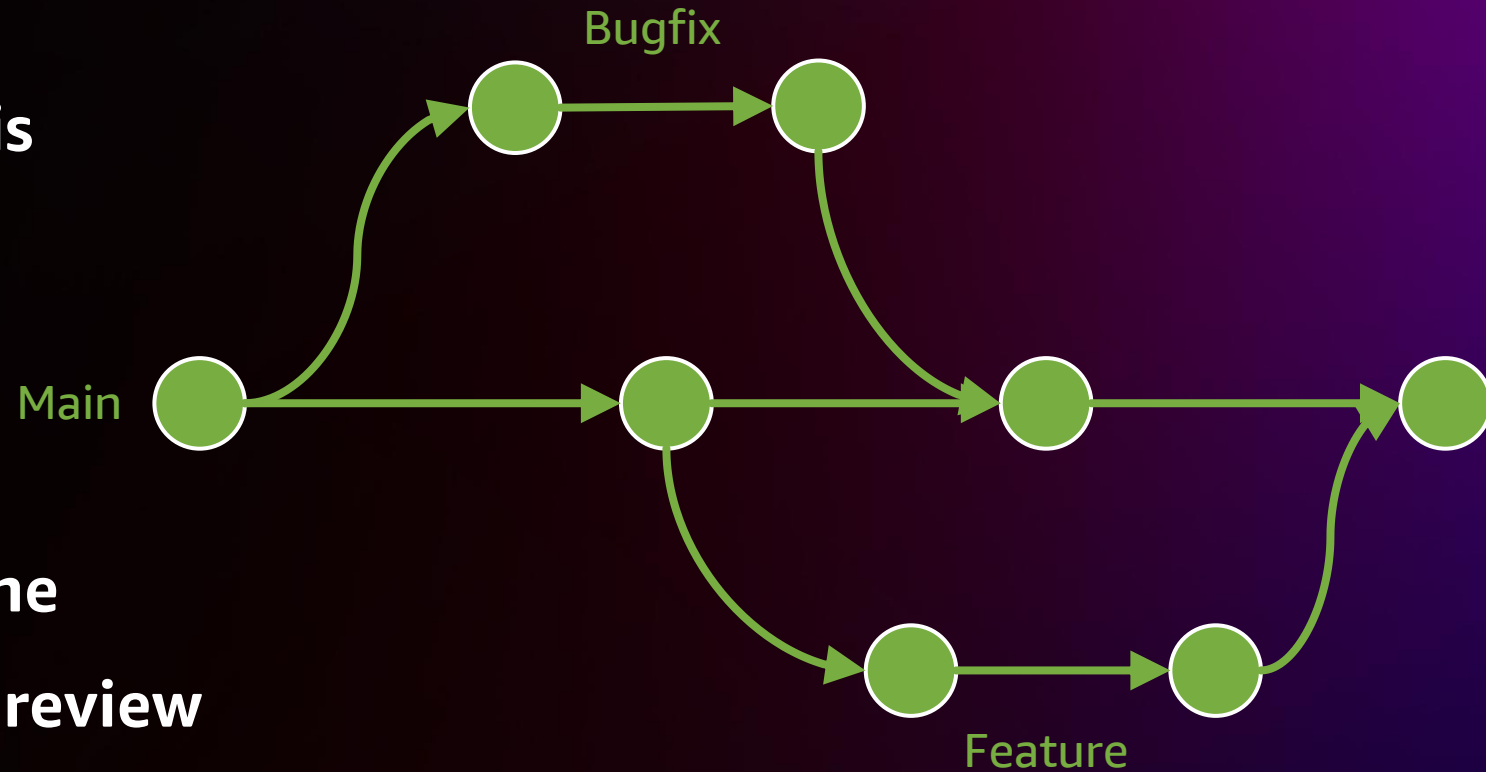
Branching strategy step 0 for CI/CD

GitFlow created by Vincent Driessen 2010!



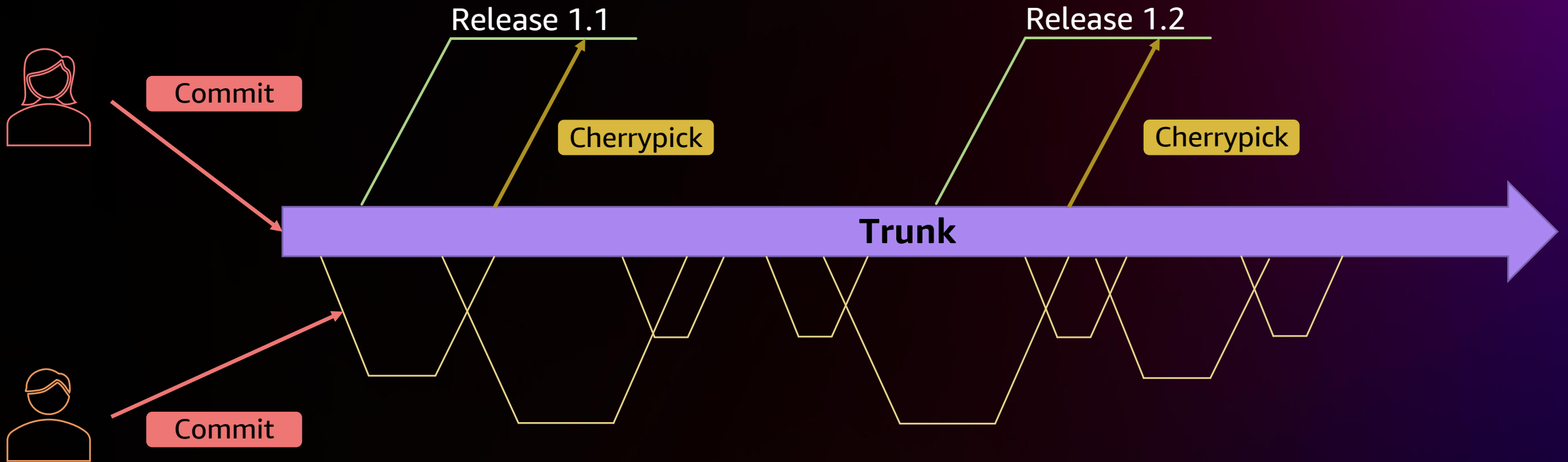
GitHub flow

1. Anything in the main branch is deployable
2. Short lived branches
3. Code review
4. Open a pull request at any time
5. Merge only after pull request review



Branching strategy: trunk-based development

- Commit in trunk at least once a day
- Trunk is always is a releasable state
- Hide unfinished code with feature toggle
- Refactor with branch by abstraction



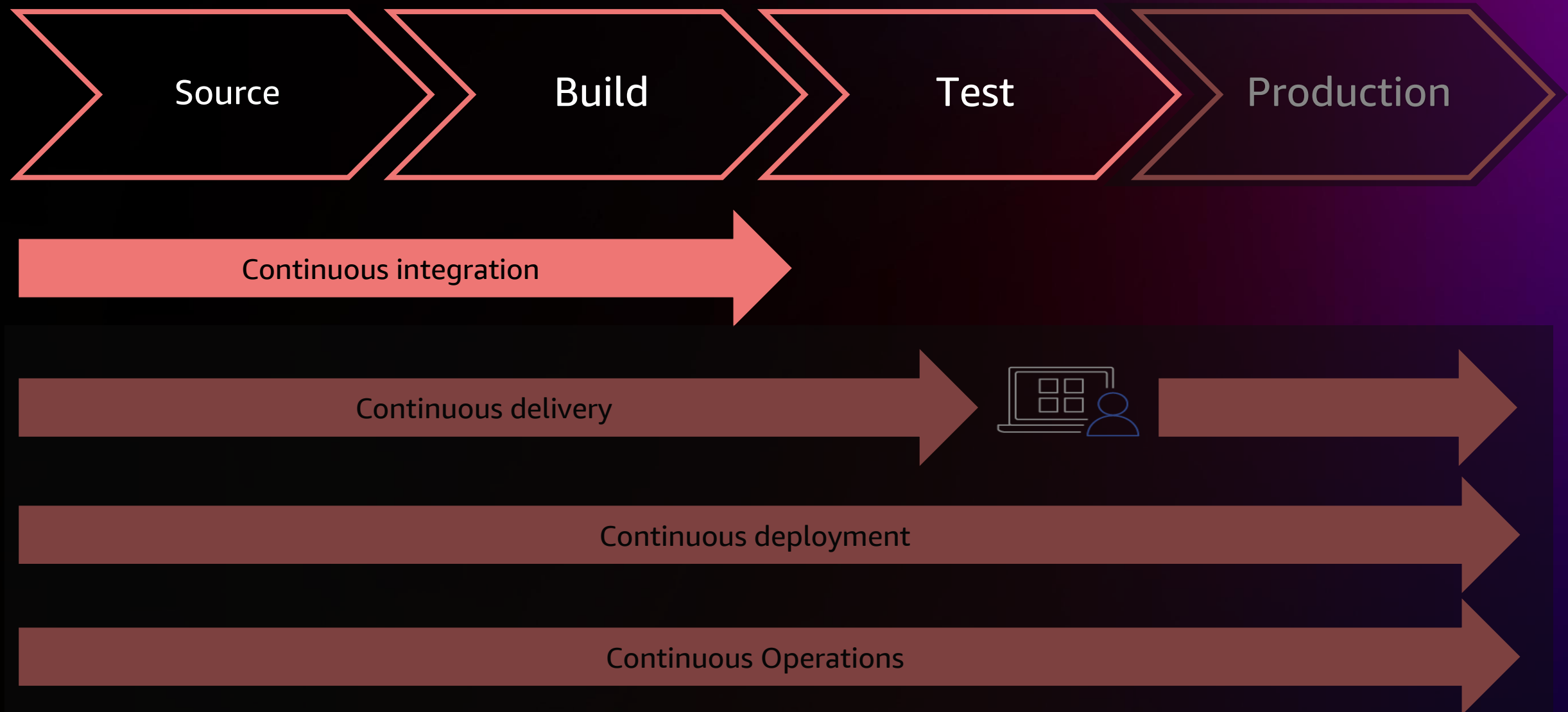
Semantic versioning 2.0.0

Given a version number MAJOR.MINOR.PATCH, increment the:

1. **MAJOR** version when you make incompatible API changes
2. **MINOR** version when you add functionality in a backwards compatible manner
3. **PATCH** version when you make backwards compatible bug fixes

PATCH version can be your **hash** for the commit in Git

Release lifecycle



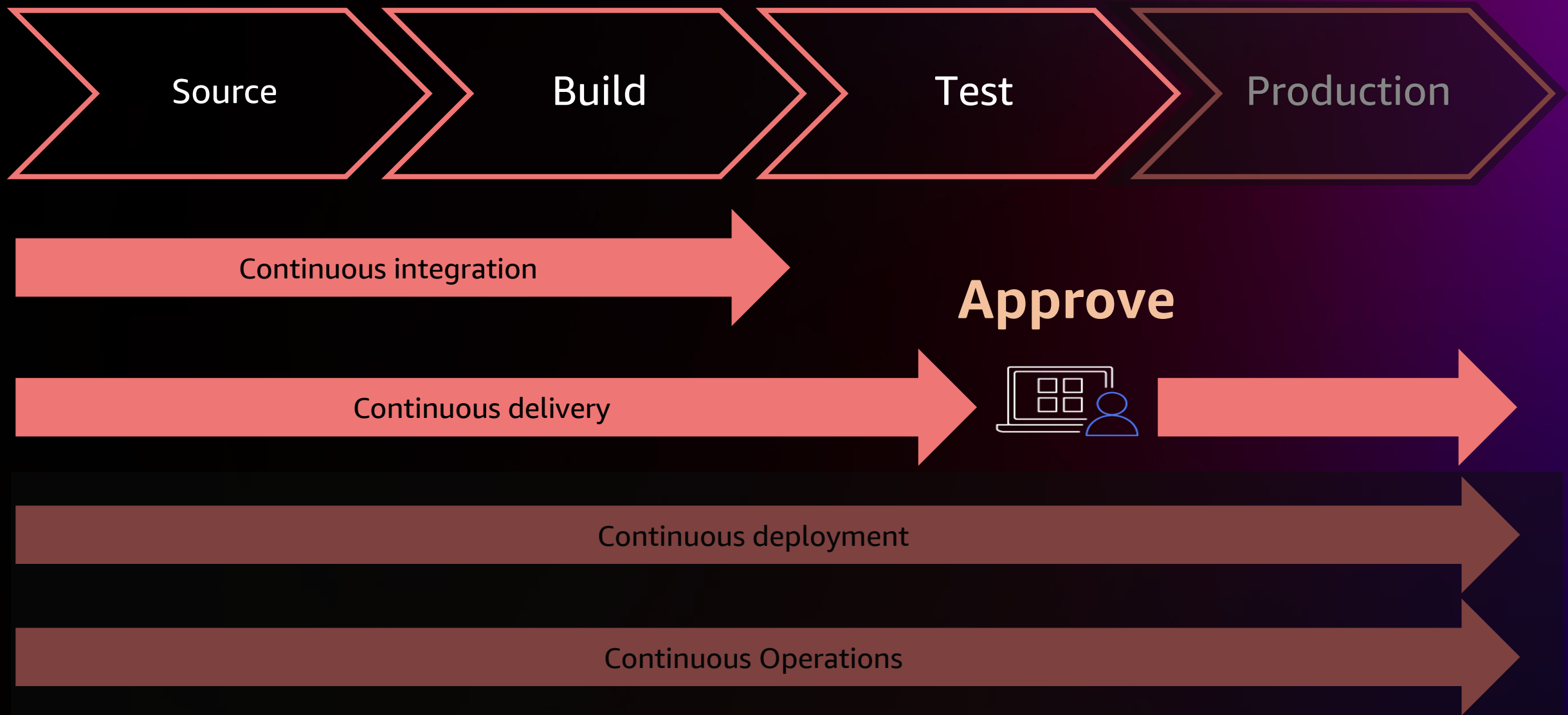
Continuous integration goals



Continuous Integration

1. **Automatically** kick off a new build when new code is checked in
2. **Build and test code** in a consistent, repeatable environment
3. Continually have an **artifact ready for deployment**
4. Continually close **feedback loop** when build fails

Release lifecycle

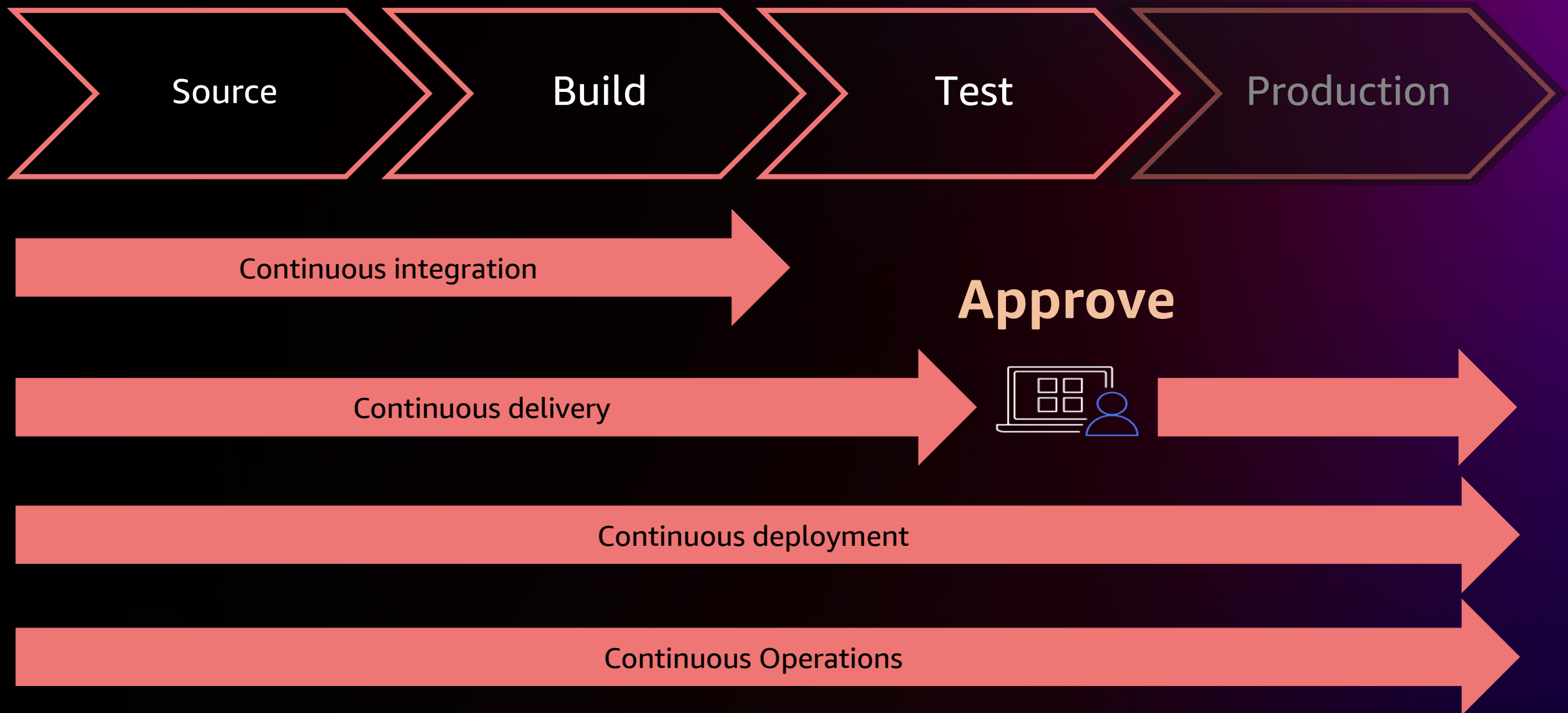


Continuous delivery

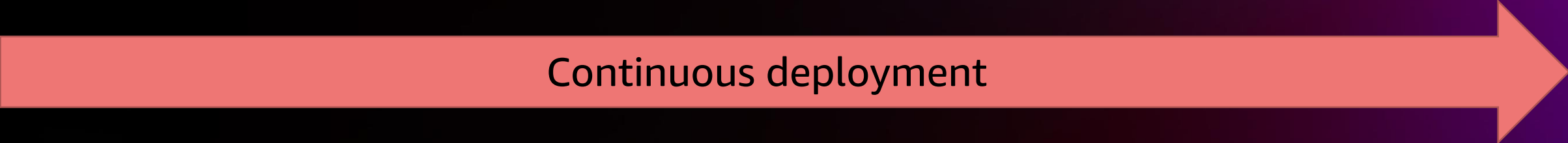
Continuous Integration +

- Automated pipeline that handles **deployment activities**
- Each change is a **release candidate**
- Automated deployment pipeline promote **build artifact**
- Production deployments triggered **manually**

Release lifecycle



Continuous deployment



Continuous deployment

1. **Automatically deploy** new changes to staging environments for testing
2. Deploy to production **safely without impacting customers**
3. Deliver to customers faster: Increase deployment frequency and **reduce change lead time and change failure rate**

Deployment strategies



Deployment strategies

Strategy	Zero Downtime	Real traffic testing	Targeted users	Cloud cost	Rollback duration	Negative impact on user	Complexity of setup
DEPLOY IN PLACE roll out new application code to an existing fleet of servers	✗	✗	✗	■ □ □	■ ■ ■	■ ■ ■	□ □ □
ROLLING version B is slowly rolled out and replacing version A	✓	✗	✗	■ □ □	■ ■ ■	■ □ □	■ □ □
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■ ■ ■	□ □ □	■ ▤ □	■ ■ □
CANARY version B is released to a subset of users, then proceed to full rollout	✓	✓	✗	■ □ □	■ □ □	■ □ □	■ ■ □
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■ □ □	■ □ □	■ □ □	■ ■ ■

Deployment strategies

Strategy	Zero Downtime	Real traffic testing	Targeted users	Cloud cost	Rollback duration	Negative impact on user	Complexity of setup
DEPLOY IN PLACE roll out new application code to an existing fleet of servers	✗	✗	✗	■ □ □	■ ■ ■	■ ■ ■	□ □ □
ROLLING version B is slowly rolled out and replacing version A	✓	✗	✗	■ □ □	■ ■ ■	■ □ □	■ □ □
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■ ■ ■	□ □ □	■ ▤ □	■ ■ □
CANARY version B is released to a subset of users, then proceed to full rollout	✓	✓	✗	■ □ □	■ □ □	■ □ □	■ ■ □
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■ □ □	■ □ □	■ □ □	■ ■ ■

Deployment strategies

Strategy	Zero Downtime	Real traffic testing	Targeted users	Cloud cost	Rollback duration	Negative impact on user	Complexity of setup
DEPLOY IN PLACE roll out new application code to an existing fleet of servers	✗	✗	✗	■ □ □	■ ■ ■	■ ■ ■	□ □ □
ROLLING version B is slowly rolled out and replacing version A	✓	✗	✗	■ □ □	■ ■ ■	■ □ □	■ □ □
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■ ■ ■	□ □ □	■ ▤ □	■ ■ □
CANARY version B is released to a subset of users, then proceed to full rollout	✓	✓	✗	■ □ □	■ □ □	■ □ □	■ ■ □
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■ □ □	■ □ □	■ □ □	■ ■ ■

Deployment strategies

Strategy	Zero Downtime	Real traffic testing	Targeted users	Cloud cost	Rollback duration	Negative impact on user	Complexity of setup
DEPLOY IN PLACE roll out new application code to an existing fleet of servers	✗	✗	✗	■ □ □	■ ■ ■	■ ■ ■	□ □ □
ROLLING version B is slowly rolled out and replacing version A	✓	✗	✗	■ □ □	■ ■ ■	■ □ □	■ □ □
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■ ■ ■	□ □ □	■ ▤ □	■ ■ □
CANARY version B is released to a subset of users, then proceed to full rollout	✓	✓	✗	■ □ □	■ □ □	■ □ □	■ ■ □
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■ □ □	■ □ □	■ □ □	■ ■ ■

Deployment strategies

Strategy	Zero Downtime	Real traffic testing	Targeted users	Cloud cost	Rollback duration	Negative impact on user	Complexity of setup
DEPLOY IN PLACE roll out new application code to an existing fleet of servers	✗	✗	✗	■ □ □	■ ■ ■	■ ■ ■	□ □ □
ROLLING version B is slowly rolled out and replacing version A	✓	✗	✗	■ □ □	■ ■ ■	■ □ □	■ □ □
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■ ■ ■	□ □ □	■ ▤ □	■ ■ □
CANARY version B is released to a subset of users, then proceed to full rollout	✓	✓	✗	■ □ □	■ □ □	■ □ □	■ ■ □
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■ □ □	■ □ □	■ □ □	■ ■ ■

Deployment strategies

Strategy	Zero Downtime	Real traffic testing	Targeted users	Cloud cost	Rollback duration	Negative impact on user	Complexity of setup
DEPLOY IN PLACE roll out new application code to an existing fleet of servers	✗	✗	✗	■ □ □	■ ■ ■	■ ■ ■	□ □ □
ROLLING version B is slowly rolled out and replacing version A	✓	✗	✗	■ □ □	■ ■ ■	■ □ □	■ □ □
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■ ■ ■	□ □ □	■ ▤ □	■ ■ □
CANARY version B is released to a subset of users, then proceed to full rollout	✓	✓	✗	■ □ □	■ □ □	■ □ □	■ ■ □
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■ □ □	■ □ □	■ □ □	■ ■ ■

How to measure your CI/CD process

DORA metrics

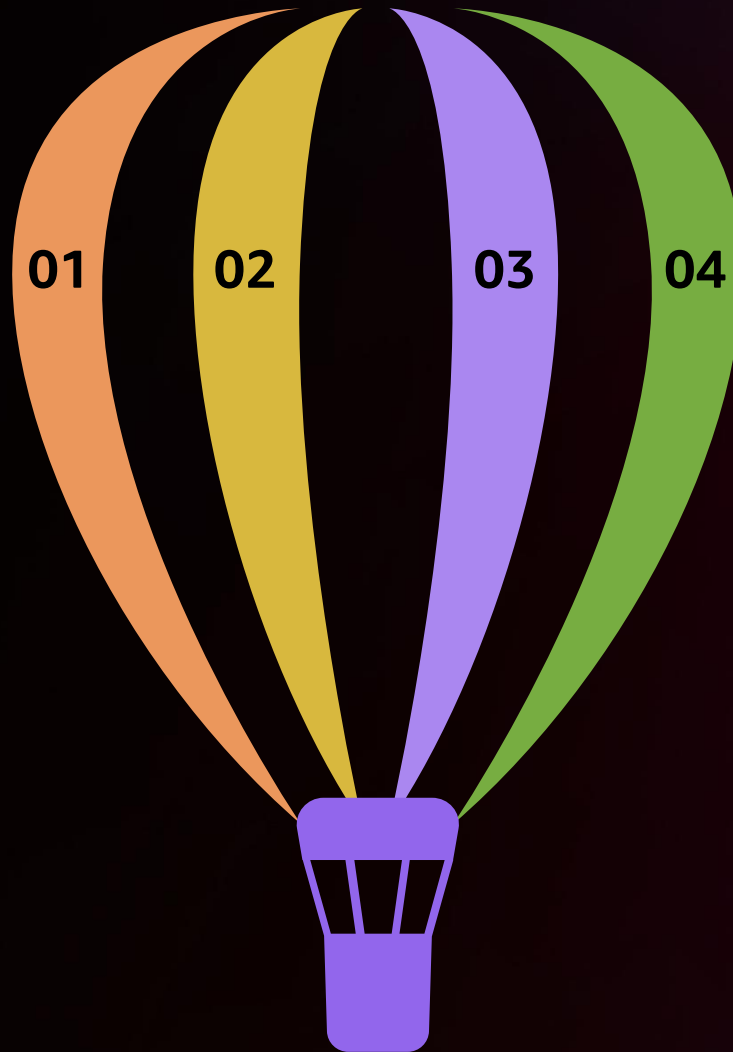
DevOps Research and Assessment (DORA) metrics

Deployment Frequency (DF)

How often an organization successfully releases to production

Lead Time for Changes (LT)

The amount of time it takes a commit to get into production



Change Failure Rate (CFR)

The percentage of deployments causing a failure in production

Time to Restore Service (TRS)

How long it takes an organization to recover from a failure in production

Effects of CI/CD

Deployment frequency	Weekly–monthly	→	On-demand
Change lead time	One–six months	→	Less than one hour
Change failure rate	16–30%	→	0–15%

26% of software teams

Source: 2021 DORA State of DevOps Report

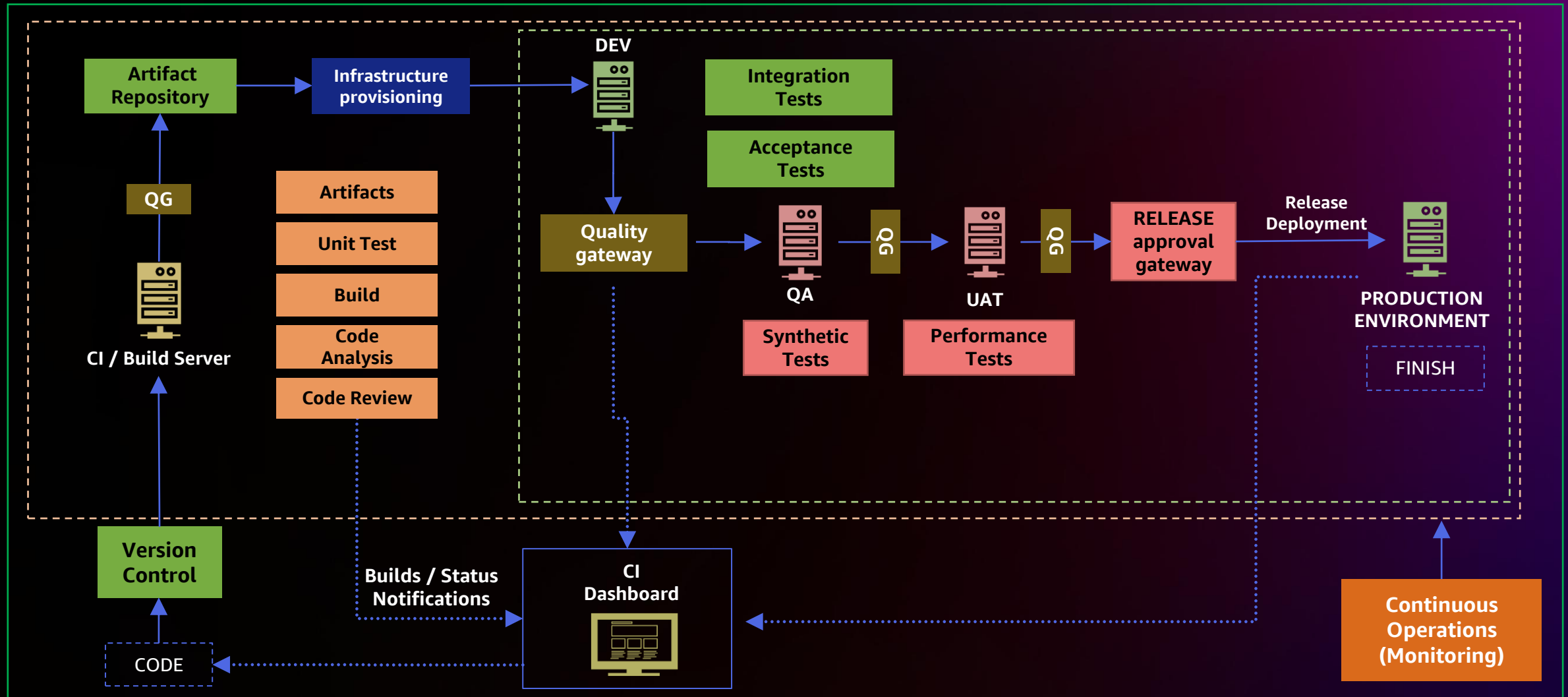
CI Metrics

Acronym	Metrics	Description	Process objectives under control	Recommended value
BFP	Build Failed Percentage	The percent of builds failed due to infrastructure reason per day vs all builds	Infrastructure problems and progress in resolving them	5-10%
T2F	Time2Feedback	Average duration of PR build	Fluctuations of feedback time over period	<30 min
PUK	Percentage of Unknown to Known	The percent of unknown errors to known errors	New infra problems (e.g., git updates)	>90%
MBSQ	Maximum BuildServer Queue	Maximal BuildServer queue on working hours	quality/performance/insufficiency of build process	<5
AQT	Average Queue Time	Avg build wait time in Jenkins queue	quality/performance/insufficiency of build process	<10m
PRC	Pull Request Cost	Calculate count of pull (or merge) request per day	if all PR created were reviewed/built in measured period	

CI/CD blueprint



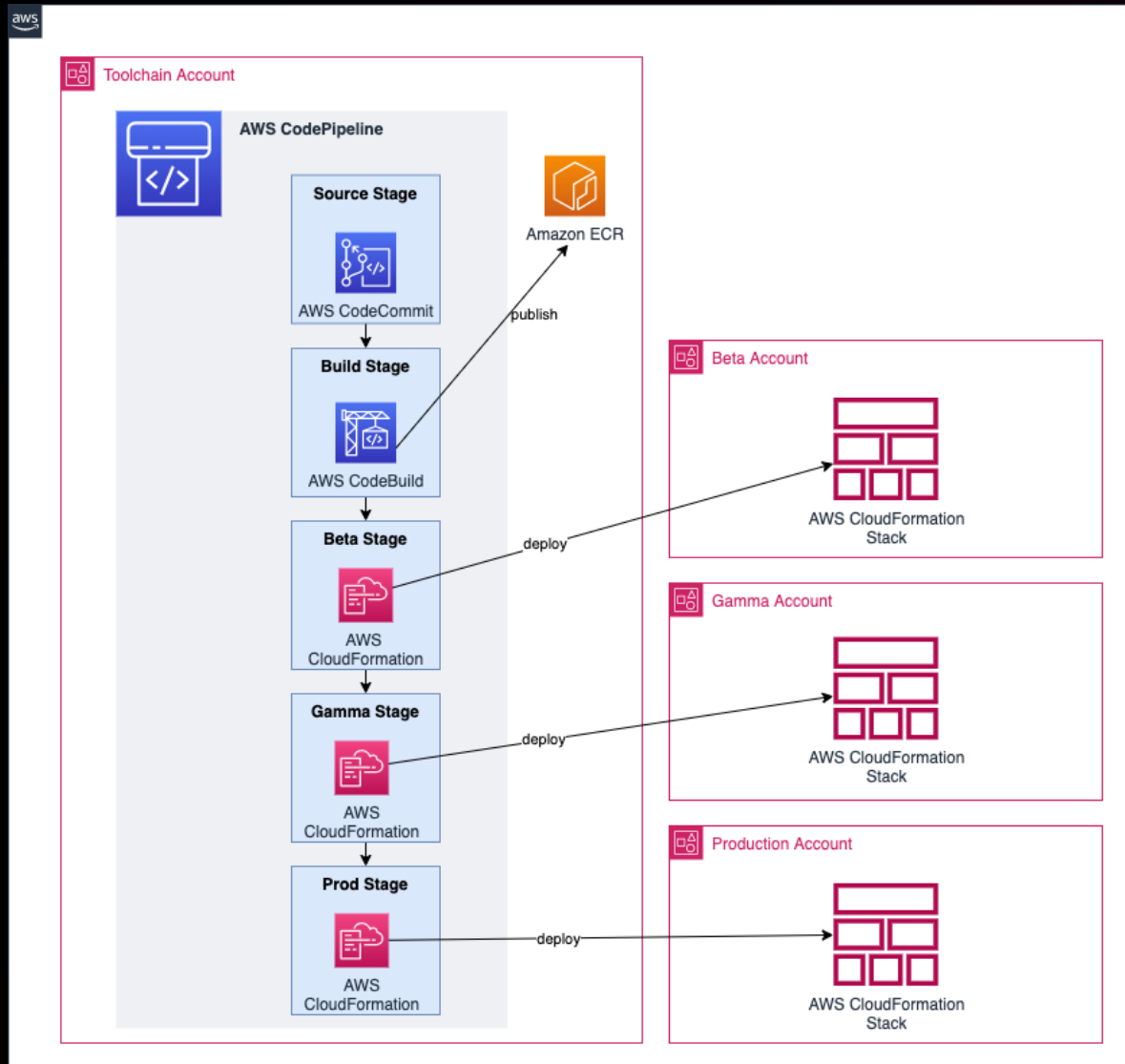
CI/CD blueprint



Reality for CI/CD pipelines



Reference implementation



Thank you!

Viktor Vedmich

 @vedmichv

Igor Ivaniuk

 igorivaniuk



Please complete the session survey in the **mobile app**



© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.